GIS Wetland Degradation & Depletion Analysis

Multi-Spectral Remote Sensing & Multi-Evidence Change Assessment Report

Run ID:	run_20260829_114614	**Baseline (T1):**	2023-08-14
Software:	GIS Engine v1.0.0	**Present (T2):**	2023-08-28
CRS:	EPSG:32633	**Interval:**	14 days
Resolution:	10.0 m × 10.0 m	**Analysis Area:**	36.00 ha

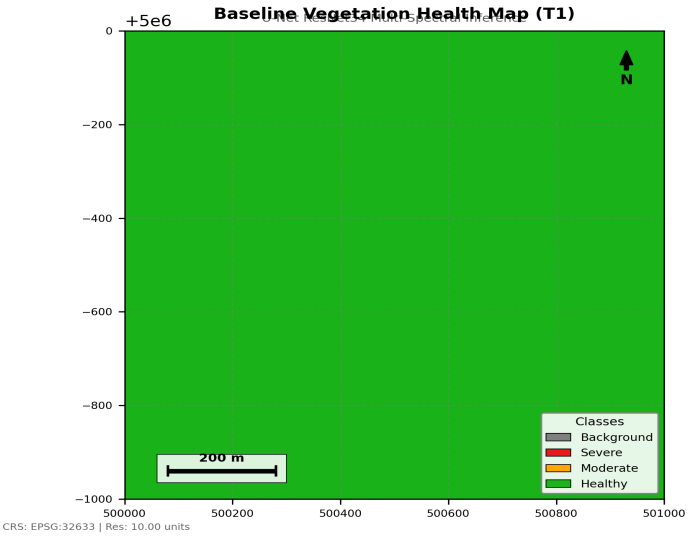
1. Executive Summary

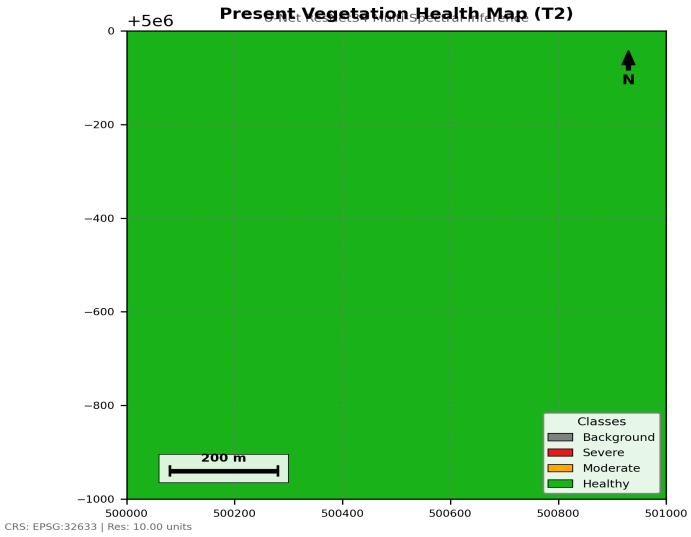
This technical assessment evaluates temporal landscape dynamics between **2023-08-14** (T1) and **2023-08-28** (T2). A total valid overlap footprint of **36.00 ha** was analyzed using a frozen 5-channel ResNet34 U-Net model, 16-state transition analysis, 3D Change Vector Analysis (CVA), and McFeeters NDWI hydrological dynamics. Primary vegetation degradation was detected across **0.00 ha (0.00%)**, while vegetation recovery was observed across **0.00 ha (0.00%)**. The Severity-Weighted Degradation footprint is **0.00 ha** (NDR: **0.0000**). **Physical Wetland Depletion Status:** NOT_ASSESSED.

Metric	Area (ha)	Percentage (%)	Classification & Scientific Scope
Vegetation Degradation	0.00	0.00%	Healthy/Mod → Severe or Healthy → Moderate
Vegetation Recovery	0.00	0.00%	Severe/Mod → Healthy or Severe → Moderate
Severity-Weighted Footprint	0.00	-	Weighted aggregate: Σ (weight × area)
Surface Water Dynamics	0.00 ha	-	NDWI surface water state shift (Loss / Gain)

2. Vegetation Health & Spatial Condition (T1 vs. T2)

Vegetation health was independently inferred for T1 and T2 using the calibrated 5-channel model (NDVI, GNDVI, NDRE, NIR, RedEdge). Four discrete classes are resolved: Background (0), Severe Stress (1), Moderate Stress (2), and Healthy Vegetation (3).

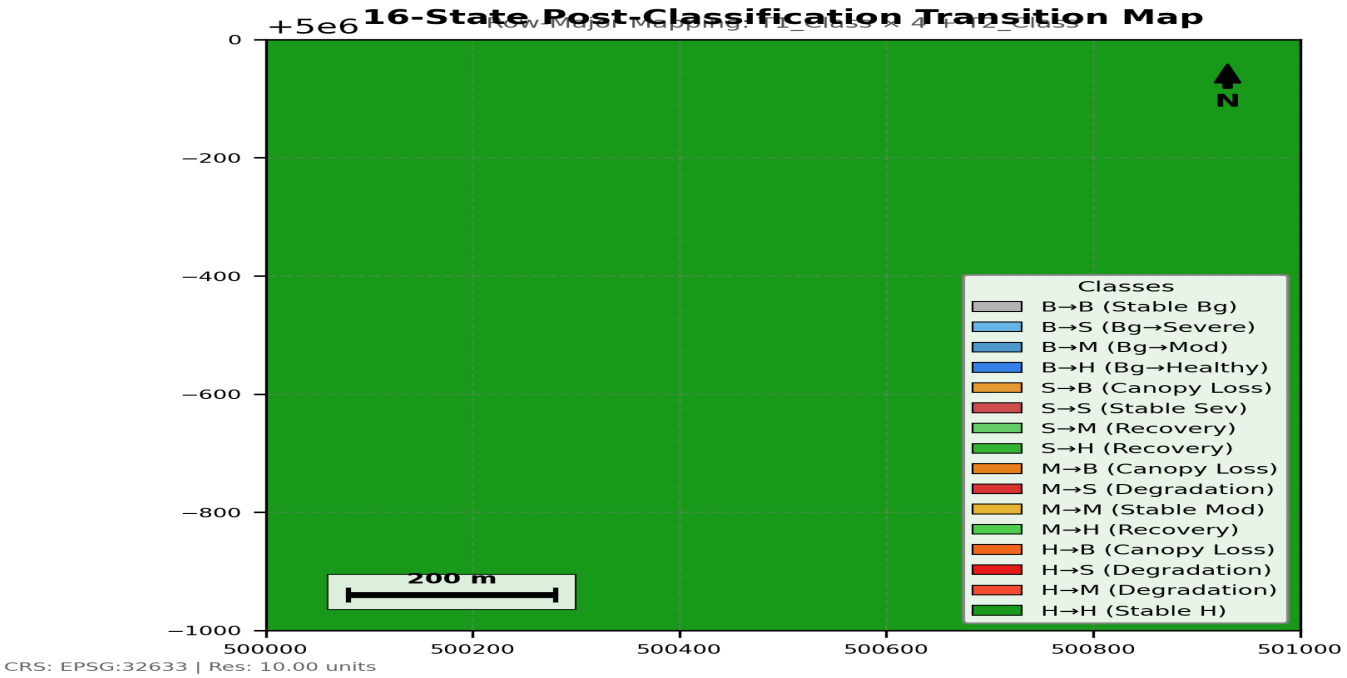




Vegetation Class	T1 Area (ha)	T1 Pct (%)	T2 Area (ha)	T2 Pct (%)	Net Shift (ha)
Healthy Vegetation	100.00	100.00%	100.00	100.00%	+0.00
Moderate Stress	0.00	0.00%	0.00	0.00%	+0.00
Severe Stress	0.00	0.00%	0.00	0.00%	+0.00
Background / Non-Veg	0.00	0.00%	0.00	0.00%	+0.00

3. 16-State Post-Classification Transition Analysis

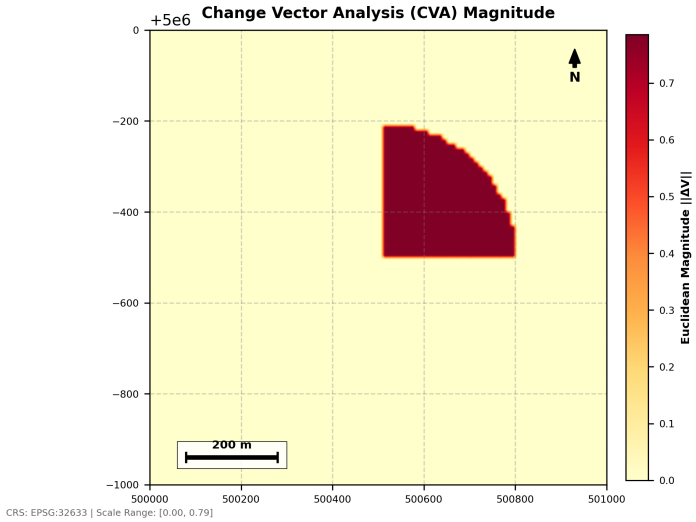
A rigorous cell-by-cell post-classification comparison maps temporal transitions using deterministic row-major indexing (T1_class × 4 + T2_class). This isolates state stability from stress progression, canopy loss, and recovery dynamics.

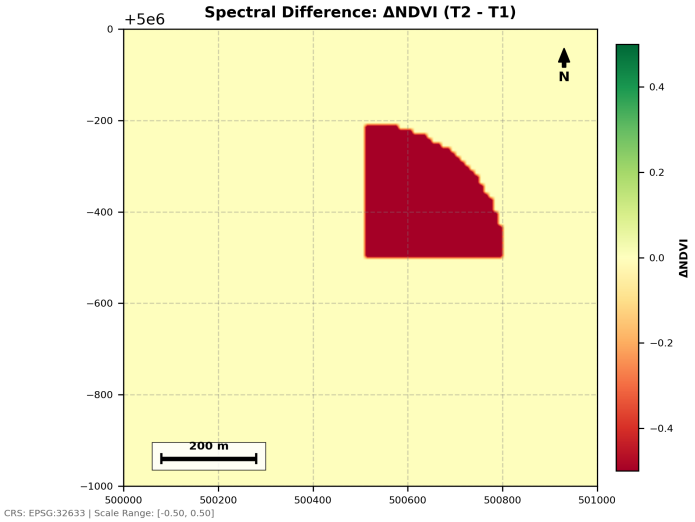


Transition Category	Transitions Included	Area (ha)	Percentage (%)
Stable State	H→H, M→M, S→S, B→B	100.00	100.00%
Vegetation Degradation	H→M, H→S, M→S	0.00	0.00%
Vegetation Recovery	M→H, S→M, S→H	0.00	0.00%
Vegetation → Background (Canopy Loss)	H→B, M→B, S→B	0.00	0.00%
Background → Vegetation (Emergence)	B→H, B→M, B→S	0.00	0.00%

4. Continuous Change Vector Analysis (CVA) & Residual Stress

Change Vector Analysis evaluates continuous multi-spectral displacement across 3D vector space ([ΔNDVI, ΔGNDVI, ΔNDRE]). CVA magnitude captures subtle intra-class stress shifts without double-counting discrete class transitions.

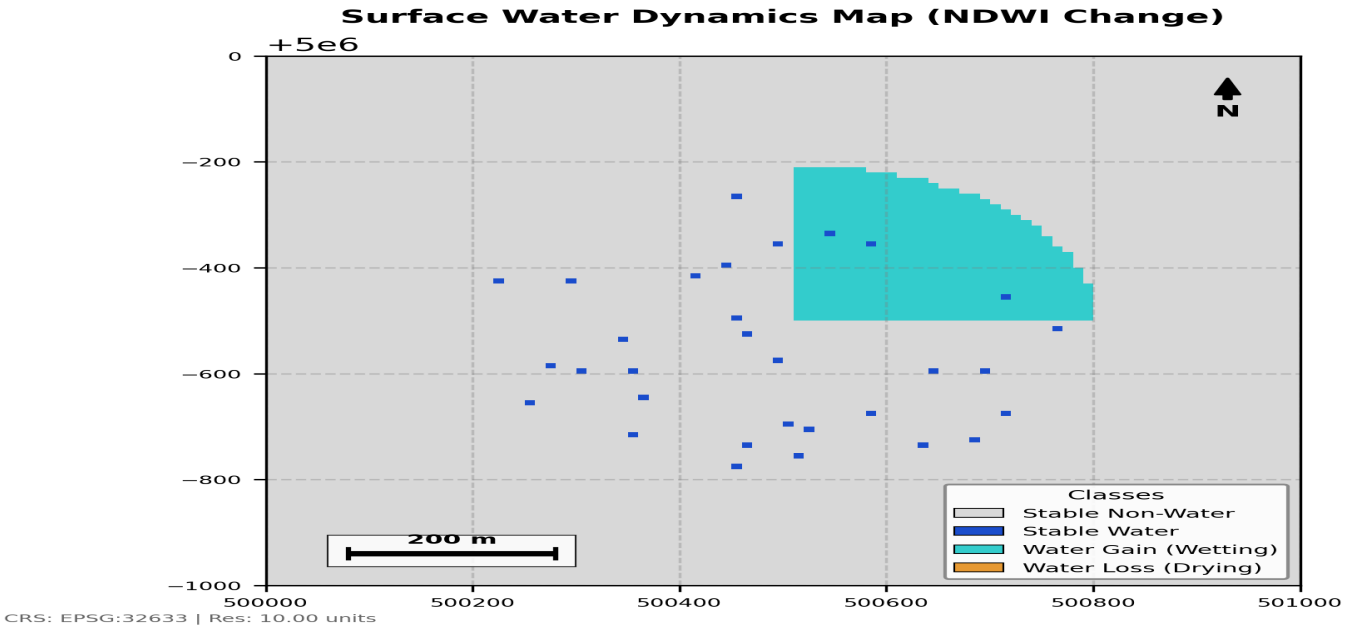




Spectral Metric	Mean Value	Median Value	95th Percentile	Physical Interpretation
CVA Magnitude (‖ΔV‖)	0.0529	0.0000	0.7837	Total spectral vector trajectory displacement
ΔNDVI (T2 - T1)	-0.0449	0.0000	0.0000	Photosynthetic vigor & chlorophyll absorption
ΔGNDVI (T2 - T1)	-0.0263	0.0000	0.0000	Green reflectance & deep canopy density
ΔNDRE (T2 - T1)	-0.0097	0.0000	0.0000	RedEdge boundary & early foliar nitrogen stress

5. Hydrological Evidence Layer (NDWI & Surface Water)

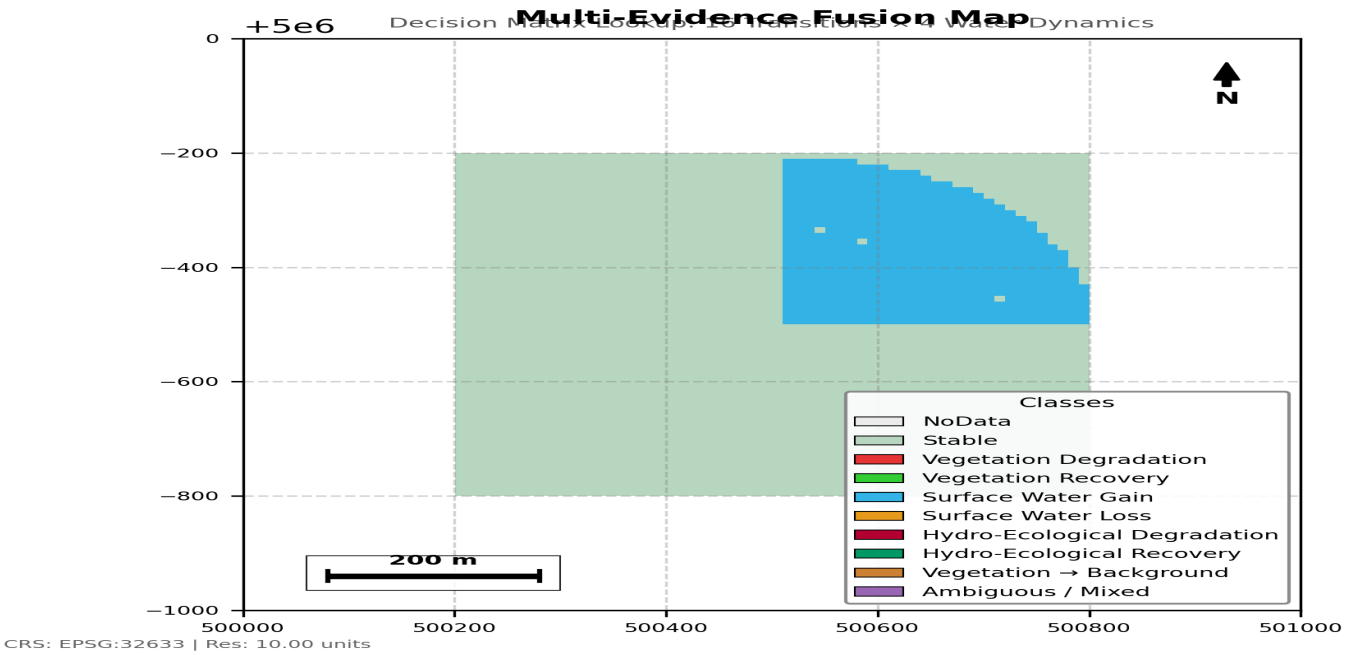
Surface water presence is modeled using the Green/NIR Normalized Difference Water Index (McFeeters 1996: [Green - NIR] / [Green + NIR]). A baseline threshold of 0.0 separates surface water from moist/vegetated soil.



Hydrological Metric	Area (ha)	Operational Status & Context
Baseline Water Extent (T1)	0.31 ha	T1 surface water area classified above NDWI threshold (2023-08-14)
Present Water Extent (T2)	7.03 ha	T2 surface water area classified above NDWI threshold (2023-08-28)
Surface Water Expansion (Gain)	6.72 ha	NDWI shifted below threshold to ≥ 0.0 (Flooding / Wetting)
Surface Water Contraction (Loss)	0.00 ha	NDWI shifted from ≥ 0.0 to below threshold (Drying / Receding)
NDWI Water Threshold	0.00	Provisional engineering threshold (Standard McFeeters zero-plane)

6. Multi-Evidence Fusion Matrix

Multi-evidence fusion integrates vegetation health transitions with surface water dynamics using an authoritative 2D decision matrix. This separates true hydro-ecological decline from dryland vegetative stress.



Fusion Evidence Class	Area (ha)	Multi-Layer Evidence Definition
Stable Landscape	100.00 ha	Persistent baseline vegetation state and moisture regime
Hydro-Ecological Degradation	0.00 ha	Concurrent Vegetation Degradation + Surface Water Loss
Hydro-Ecological Recovery	0.00 ha	Concurrent Vegetation Recovery + Surface Water Gain
Vegetation-Only Degradation	0.00	Vegetation health decline occurring over stable water/moisture regime
Vegetation-to-Background (Canopy Loss)	0.00 ha	Vegetation loss without independent physical depletion confirmation

7. Quality Assurance, Uncertainty & Production Validation

All analytical steps are checked by the automated Quality Assurance engine to enforce mathematical reconciliation, spatial grid integrity, and conservative scientific bounds.

Validation Check / Gate	Status	Quality Assurance Finding
Input Geometry & CRS Alignment	PASS	Exact grid match across dimensions, affine transform, and resolution.
Mathematical Area Reconciliation	PASS	Transition matrix sum matches valid overlap footprint (< 1e-6 ha residual).
Spectral CVA Distribution Check	PASS	No NaN/Inf leakage; finite bounds verified across all spectral bands.
Independent Ground Reference Data	UNAVAILABLE	Independent ground-truth samples not configured. Areas are model estimates.
Physical Depletion Claim Guard	ENFORCED	Safeguard active: Vegetation decline is not labeled physical wetland loss.

8. Scientific Synthesis, Interpretation & Limitations

OBSERVED (Direct Measurements): Pixel spectral reflectance changes (ΔNDVI, ΔGNDVI, ΔNDRE), CVA vector magnitude, and NDWI water plane shifts.
INFERRED (Multi-Layer Evidence): Vegetation health condition classes, canopy degradation/recovery transitions, and multi-evidence hydro-ecological stress.
NOT VALIDATED (Uncertainty Notice): Physical wetland loss/depletion cannot be confirmed without multi-year ground-truth validation or official wetland boundary records.

Operational Limitations & Technical Notice:

- **Independent Reference Data:** Ground reference validation samples are currently unavailable. Reported areas represent model-derived estimates rather than error-adjusted Olofsson confidence intervals.
- **Provisional Severity Weights:** Severity weighting factors (Healthy=0.0, Moderate=0.35, Severe=0.80, Background=1.0) are engineering calibration parameters and not field-calibrated ecological constants.
- **Spectral Bandpass Availability:** Drone/Sentinel 4-band data lacks SWIR; hence, MNDWI and LSWI formulations are excluded.
- **Phenological Influence:** Temporal gaps between acquisitions may reflect seasonal senescence rather than permanent anthropogenic degradation.